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Class Learning Activity

Instructions to answer:

1. Define what research is. How about the Capstone Project

- ***Research*** is a structured process of gathering, analyzing, and interpreting data to answer questions, solve problems, or generate new knowledge. It relies on evidence and logical reasoning to ensure accuracy and validity. A ***Capstone Project***, meanwhile, is a culminating academic requirement where students apply what they have learned throughout their program to address real-world problems or create innovative solutions, often blending research with practical application.

2. Classify and differentiate the different types of research.

- Research can be classified into types based on approach and purpose. ***Quantitative*** research deals with numbers, statistics, and measurable data, while ***qualitative*** research focuses on descriptions, experiences, and meanings. Applied research aims for practical problem-solving, whereas basic research seeks to expand general knowledge. ***Descriptive*** research observes and records facts as they occur, while experimental research tests hypotheses under controlled conditions. Their differences lie mainly in their goals, methods, and expected results.

3. What are the purpose and characteristics of research?

- The ***purpose of research*** is to discover new knowledge, verify facts, and solve problems, which in turn supports decision-making and innovation. Its main characteristics include being systematic, empirical, logical, replicable, and objective. These ensure that research follows a clear process, relies on evidence, uses sound reasoning, can be repeated by others, and avoids bias for reliable results.

4. Define what research methodology is and its five types?

- ***Research methodology*** refers to the overall strategy, design, and process used in conducting a study. It encompasses the techniques, tools, and procedures for data collection and analysis, ensuring the research is systematic and valid. Five common types of research methodology include:
    1. **Quantitative Methodology** – focuses on numerical data and statistical analysis.
    2. **Qualitative Methodology** – emphasizes descriptive, non-numerical insights.
    3. **Mixed-Methods Methodology** – combines quantitative and qualitative approaches for a more comprehensive understanding.
    4. **Experimental Methodology** – involves controlled testing of hypotheses.

5. **Descriptive Methodology** – documents and describes phenomena without manipulating variables.

5. What is the meaning of Design?

- ***Design*** is the process of planning and creating a concept, system, or product to meet specific needs or solve problems. It requires creativity, organization, and purpose, often blending functionality with aesthetics. In research, design also refers to structuring the study to ensure data is collected and analyzed effectively.

6. How are Research and Design is distinguished according to Ashley Karr?

- ***Ashley Karr*** explains that research focuses on understanding "**what is**" by exploring facts, patterns, and truths, while design focuses on creating "**what could be**" through innovative solutions. Research analyzes and explains reality, while design imagines and builds possibilities. Both are different but can work together.

7. What is the Gray between Design and Research? Illustrate them.

- The "**gray area**" between research and design is where both overlap and influence each other. Research can use design thinking to plan studies, and design can rely on research to make informed decisions. For example, in product development, user research shapes the design, and prototypes are improved through further testing. This creates a continuous cycle of improvement.

8. What's the difference between the Research versus the Design Problem

- A ***research problem*** identifies a gap in knowledge or understanding that needs investigation, asking, "**What do we not know, and how can we find out?**" A ***design problem*** focuses on creating or improving something to meet specific needs, asking, "**What can we create or change to solve this issue?**" Research problems aim to explain or discover, while design problems aim to invent or improve.